LING10 Statistics for Linguistics
Dartmouth College. Winter 2026

Homework 4
Interactions, Model Selection, Logistic Regression and Random Effects

(1) Iconicity, sensory experience and parts-of-speech in English

Words can have a property called `iconicity`. Some spoken words are iconic, exhibiting a resemblance between form and meaning. For instance, the word *boom* bears resemblance to the loud, explosive sound to which it refers, and the word *teeny* conveys a sense of smallness through the high-front vowel /i/.

The `SER`, or *sensory experience rating*, is a measurement of whether a word "evokes a sensory experience" or not. It is a scale from 1 to 7. When the data was compiled, the experiment's instructions emphasized the five common senses: taste, touch, sight, sound and smell. The word with the highest sensory experience rating was *garlic* (6.56), followed by *walnut* (6.5) and *water* (6.33). The lowest sensory experience rating (1.0) was shared between many words, including *an*, *for* and *hence*. Words rated low in sensory experience were mostly function words, but also included some nouns, such as *choice* (1.0), *guide* (1.09) and *bane* (1.10).

What kinds of words are more likely to be iconic? Is It words that can be connected to senses? Is it verbs (which can be directly connected to movements and actions)? The dataset `pos-iconicity-100.csv` has information for 100 words. Use this data to study the interaction of `SER` and `POS` {noun, verb} when predicting `Iconicity`.

(a) Does this model meet the assumptions of a linear model? (Assume the residuals are normal and that variables are not collinear).

(b) Write the equation of the model for the linear regression `Iconicity ~ SER*POS`

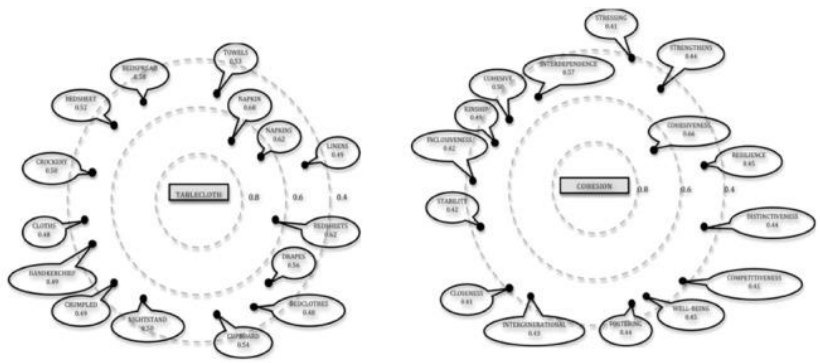
(c) Write a short but formal report for the linear regression result, as if you were writing a conference abstract. What does the interaction mean? Include a chart with your explanation. You can use the following as the first line of your ggplot chart:

```
ggplot(ico2, aes(x=SER, y=Iconicity, group=POS, color=POS))
```

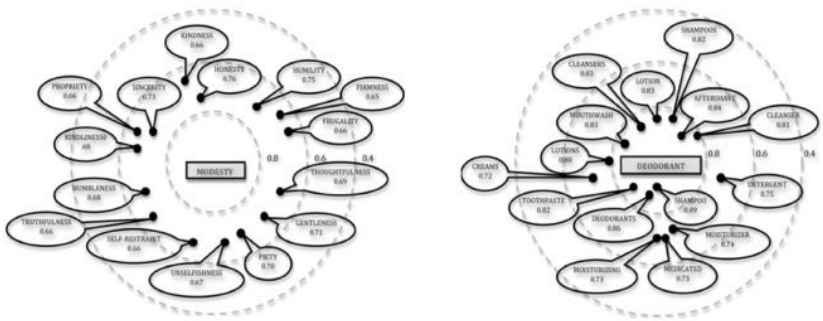
The data is real: Sidhu & Pexman (2018). *Lonely sensational icons: semantic neighbourhood density, sensory experience and iconicity*. Language, Cognition and Neuroscience 33(1):25-31.

(2) Iconicity, sensory experience and semantic neighborhoods

Let's continue our examination of iconicity. Words can have *semantic neighborhoods*. A word has a *dense* semantic neighborhood if there are many other words that are similar or related in meaning. A word has a *sparse* semantic neighborhood if there are very few words that are similar or related in meaning. The figure below illustrates this with a few examples.



Low SND: Neighbours are less related to the target word (on average)



High SND: Neighbours are more related to the target word (on average)

The dataset `iconocity_50.csv` has 100 words and their values for `Iconicity` and sensory experience rating (`SER`). It also has the continuous variable `ARC` (*average ratio of co-occurrence*). This is a measurement of semantic neighborhood density. A word with a low ARC is in a sparse semantic neighborhood. A word with a high ARC is in a dense semantic neighborhood.

(a) Does the model meet the assumptions of a linear model? (Assume that the residuals are normally distributed and that the variables are not collinear).

(b) Write the equation for the model in the linear regression `Iconicity ~ SER*ARC`

(c) Write a short but formal report for the linear regression result. What does the interaction mean? Include a chart with your explanation. I recommend that you split the ARC variable in half, using the code below:

```
summary(ico$ARC)
ico$highLowARC = "LowARC: SparseNeighborhood"
ico$highLowARC[ico$ARC > 0.6] = "HighARC: DenseNeighborhood"

ggplot(ico, aes(x=SER, y=Iconicity)) +
  facet_wrap(~highLowARC)+
  geom_point() +
  geom_smooth(method="lm")+
  scale_y_continuous(name = "Iconicity")+
  scale_x_continuous(name = "SER") +
  ggtitle("Iconocity by SER and ARC")
```

(3) Tongue positions that are difficult to measure

The figure below shows the shape of the tongue as people pronounce different sounds in English. These charts were produced by draping 15 small gold pellets over the tongue. However, the pellets at the back of the throat are very difficult to place. Johnson, Ladefoged & Lindau (1993) tried to figure out if there was a way to predict the x and y positions of the 15-th pellet (`x15`, `y15`) using the positions of the pellets in the front of the mouth. (The frontmost pellet is #1, the backmost pellet is #15. The #15th pellets are marked with an arrow).

The dataset `chaindata.csv` contains information collected for 54 vowels and for 30 variables (the x and y positions of 15 sensors). Can the position of `y15` (the y-position of the backmost pellet) be predicted from these variables? Try the initial model:

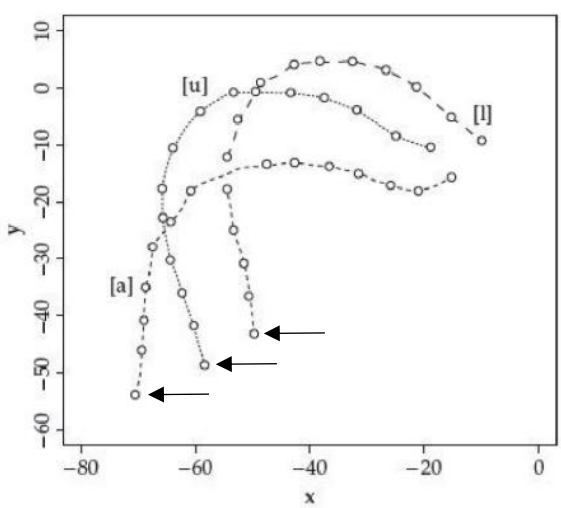
```
y15  ~ x1 + y1 + x2 + y2 + x4 + x5 + y5 + x6 + y6
```

(a) Run diagnostics on the initial model (normality of residuals, non-variance of residuals, collinearity, autocorrelation). Does it show any problems? (Assume that the autocorrelation is not a problem).

(b) Try to run the stepAIC instruction in three directions: forward, backward and "both". What are the simplest models that can describe `y15`? Do the results of all three directions match?

(c) Use the simplest suggestion out of all of the stepAIC suggestions. Does this reduce the AIC? (Calculate the AIC of this simpler model with the `AIC(modelName)` instruction. Don't use the AIC number from the stepAIC directly). What is the adjusted R^2 of this simpler model, compared to the adjusted R^2 of the initial model?

(d) Let's think about the differences between the initial model and the simplest model. Which variables are significant in the initial model? Which ones are significant in the simplest model? How is the simplest model different from the first one?



This is real data: Johnson, K., Ladefoged, P., Lindau, M. (1993). *Individual Differences in Vowel Production*. J. Acoust. Soc. Am. 94(2).

(4) Logistic Regression, Back to Panama

Cedergren (1973) noticed that speakers in Panama City, like in many dialects of Spanish, variably deleted the /s/ at the end of words. She carried out a study to find out if /s/ was systematically dropping out of Panamanian Spanish. She also was interested in the effect of social constraints on final /s/ deletion. Cedergren recorded 8846 words whose standard Spanish variant has final /s/. For each token she recorded whether the word (`deletion`), as well as the social class of the speakers (`class`). The data is in the dataset `big-panama.csv`. The structure of the variables is as follows:

`Deletion`: whether there was a pronounced final /s/ on the word
1=deletion
0=non-deletion

`Class`: the "social class" of the utterer
1=highest
2=second highest
3=second lowest
4=lowest

What is the relationship between social class and /s/ deletion? Your answer should have four parts:

(1) The equation for your generalized linear model. Remember to include your error term and the link function.

(2) A formal report of a generalized Linear Model with `class` as an independent variable and `deletion` as the dependent variable. Remember that the number 1 in `class` is the highest class, not the lowest.

(3) Calculate the concordance index of this model. Do you think this provides sufficient discrimination? Should we add more variables?

(4) Use the `predict` instruction to get the predicted percentage of deleted /s/ for each social group.

The data is real. It comes from: Cedergreen, Henrietta (1973). The interplay of social and linguistic factors in Panama. Dissertation: Cornell University.

(5) Random Effects: Tones make you sad

The dataset `happy-tones.csv` has a combination of linguistics and social databases. Most of the dataset comes from WALS, such as the language identifiers. The variable `tones` tells you how many tones a language has. The variable `happyPlanetIndex` is calculated according to the reported happiness of a country and the sustainability of its economy (more information here). Is speaking tonal languages correlated to overall happiness? You need to do three things:

(1) Make a chart for the relationship between `tones` (x-axis) and `happyPlanetIndex` (y-axis).

(2) Run a linear mixed-effects model, with `tone` as the independent variable, `happyPlanetIndex` as the dependent variable, and `family` and `macroarea` as random effects with a random intercept (e.g. `(1|family) + (1|macroarea)`). Write a formal report and a 1~2 sentence summary of the results without using any numbers.

(3) Run a linear mixed-effects model, with `tone` as the independent variable, `happyPlanetIndex` as the dependent variable, and `family` and `macroarea` as random effects with a random intercept and a random slope for tone (e.g. `(1+tone|family) + (1+tone|macroarea)`). Write a formal report and a 1~2 sentence summary of the results without using any numbers.

(4) Make a chart for the relationship between `tones` (x-axis) and `happyPlanetIndex` (y-axis), but using different slopes for `macroarea`. Using the information on this chart and the two models, write a simple 3~5 sentence response to the following question: What are the slopes telling you? What is the difference between the first and the second model?

The data is real. The tone data comes from WALS (https://wals.info/feature/13A#2/19.3/153.1) and the happiness data comes from Happy Planet (https://happyplanetindex.org/).

Evaluation

You need to turn in a Word/LibreOffice document with the answers to each of the questions. You also need to turn the R code that you used to get to your results.

20% x 5 Five questions

This homework is due at 11:59 pm of **Tuesday, March 10th, 2026**.